

Shiny
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PROJECT HORIZON

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- Module 1 - Data Generation
- Module 2 - Data Preprocessing
- Module 3 - Drone Swarm Simulation
- Module 4 - Multi-Agent Reinforcement Learning
- Module 5 - Performance Evaluation

PROJECT HORIZON

Cooperative Drone Swarm Search and Rescue
Multi-Agent Reinforcement Learning and Swarm-Based Agent Simulation

Project Modules

1. Module 1 - Data Generation, Statistical Analysis and Visualization
2. Module 2 - Data Preprocessing
3. Module 3 - Multi-Agent Drone Swarm Simulation
4. Module 4 - Multi-Agent Reinforcement Learning
5. Module 5 - Performance Evaluation

System Workflow

Dataset Generation → Preprocessing → Drone Swarm Simulation → MARL Training → Performance Evaluation

Module 1 - Data Generation, Statistical Analysis and Visualization

Dataset Statistical Analysis Visualization

Dataset Generation Method:

GMM ▼

Number of Records:

500 1,000

500 600 700 800 900 1,000

GENERATE DATASET

Download CSV

5006007008009001,000

GENERATE DATASET

Download CSV

Show10entries

Search:

Drone_ID	Battery	Speed	Altitude	Latitude	Longitude	Signal	Target_Distance	Obstacle_Distance	Reward	Status
All	All	All	All	All	All	All	All	All	All	All
1	75.18	30.03	230.8	17.582	79.4675	87.88	31.06	8.93	1.66	Target Detected
2	45.26	60.95	446.67	15.5463	78.9331	92.75	32.45	23.59	4.28	Searching
3	51.04	40.5	306	12.4785	80.4531	54.55	87.03	9.41	8.25	Target Detected
4	39.45	69.42	457.57	17.0701	77.3537	73.57	32.87	46.23	9.91	Target Detected
5	46.18	71.23	428.42	12.1787	80.4933	94.29	12.57	7.91	1.69	Searching
6	74.12	34.34	209.3	16.888	80.9492	40.49	35.62	22.55	5.85	Returning
7	49.6	53.71	338.69	14.3445	76.5973	49.81	93.07	41.22	4.05	Returning
8	39.24	65.81	440.34	12.8604	77.6552	55.59	87.52	22.28	8.33	Searching
9	71.15	58.85	349.47	15.9993	76.1283	86.37	82.02	10.05	7	Idle
10	61.39	57.87	277.66	15.7115	79.5775	44.27	2.18	12.19	1.1	Searching

Showing 1 to 10 of 1,000 entries

Previous12345...100Next

Dataset

Statistical Analysis

Visualization

Descriptive Statistics

Variable	Mean	Median	SD	Variance
Drone_ID	500.50	500.50	288.82	83416.67
Battery	63.12	64.93	18.09	327.15
Speed	50.41	50.86	15.09	227.73
Altitude	298.32	281.76	103.08	10625.46
Latitude	14.99	14.99	1.73	2.98
Longitude	78.94	78.95	1.70	2.89
Signal	69.96	69.37	17.20	295.72
Target_Distance	50.23	49.67	28.25	798.19
Obstacle_Distance	27.76	28.08	13.22	174.68
Reward	4.50	4.42	3.20	10.22

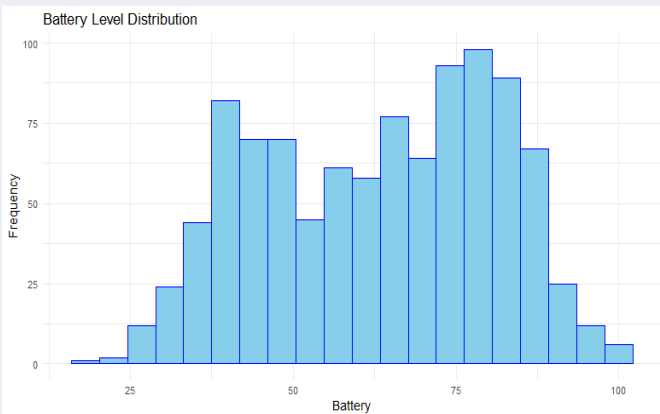
Missing Value Analysis

Variable	Missing	Percentage
Drone_ID	0	0.00
Battery	0	0.00
Speed	0	0.00
Altitude	0	0.00
Latitude	0	0.00
Longitude	0	0.00
Signal	0	0.00
Target_Distance	0	0.00
Obstacle_Distance	0	0.00
Reward	0	0.00
Status	0	0.00

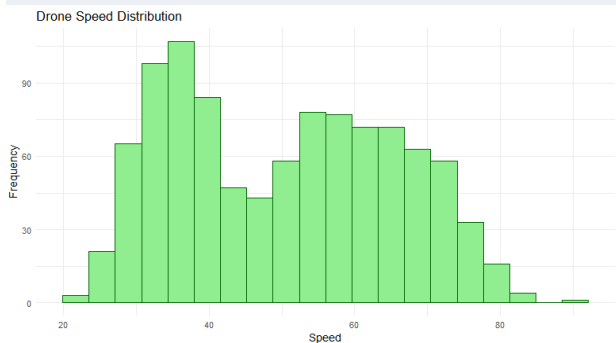
Correlation Matrix

Drone_ID	Battery	Speed	Altitude	Latitude	Longitude	Signal	Target_Distance	Obstacle_Distance	Reward
1.00	-0.01	0.01	0.01	-0.01	0.01	0.02	-0.01	0.01	-0.01
-0.01	1.00	-0.81	-0.83	0.01	-0.01	-0.02	-0.00	0.00	-0.02
0.01	-0.81	1.00	0.85	-0.04	0.01	0.02	-0.02	-0.01	0.01
0.01	-0.83	0.85	1.00	-0.04	-0.00	0.01	-0.01	-0.00	-0.00
-0.01	0.01	-0.04	-0.04	1.00	-0.01	0.00	0.04	0.05	0.00
0.01	-0.01	0.01	-0.00	-0.01	1.00	-0.06	0.00	0.02	0.00
0.02	-0.02	0.02	0.01	0.00	-0.06	1.00	-0.02	0.00	-0.02
-0.01	-0.00	-0.02	-0.01	0.04	0.00	-0.02	1.00	-0.02	-0.03
0.01	0.00	-0.01	-0.00	0.05	0.02	0.00	-0.02	1.00	-0.02
-0.01	-0.02	0.01	-0.00	0.00	0.00	-0.02	-0.03	-0.02	1.00

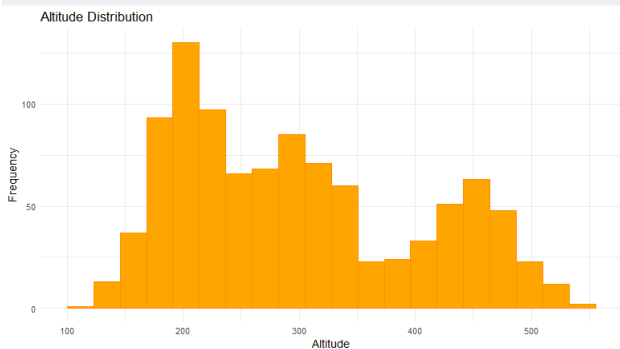
1. Battery Level Distribution



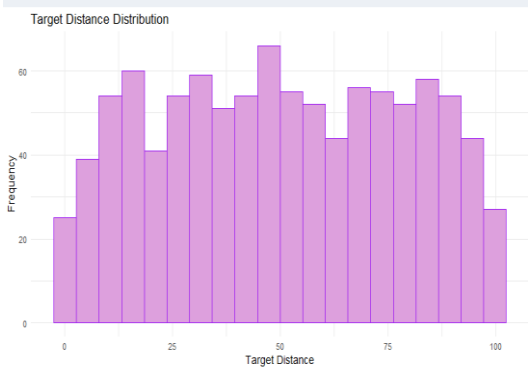
2. Drone Speed Distribution



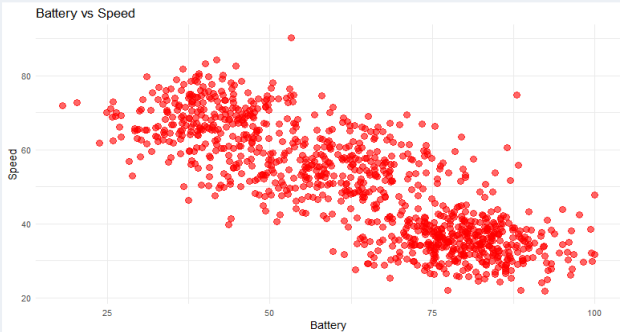
3. Altitude Distribution



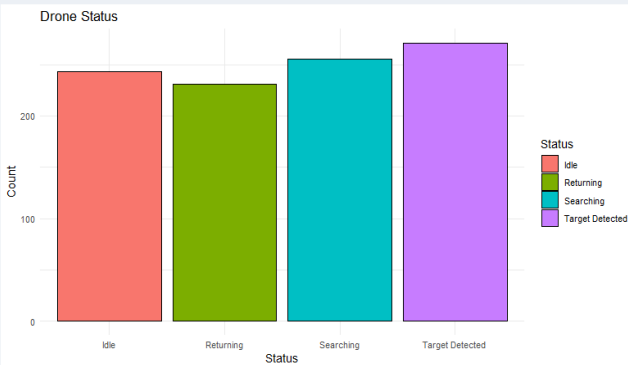
4. Target Distance Distribution



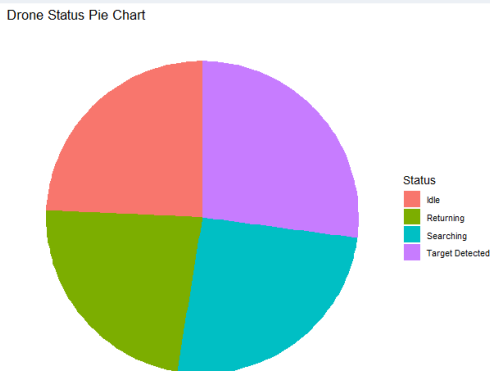
5. Battery vs Speed



6. Drone Status



7. Drone Status Pie Chart



Module 2 - Data Preprocessing

Raw Data

Missing Values

Process Data

Before vs After

The dataset generated in Module 1 is used as input for preprocessing.

Show10entries

Search:

Drone_ID	Battery	Speed	Altitude	Latitude	Longitude	Signal	Target_Distance	Obstacle_Distance	Reward	Status
All	All	All	All	All	All	All	All	All	All	All
1	75.18	30.03	230.8	17.582	79.4675	87.88	31.06	8.93	1.86	Target Detected
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5	46.18	71.23	428.42	12.1787	80.4933	94.29	12.57	7.91	1.89	Searching
6	74.12	34.34	209.3	16.888	80.9492	40.49	35.62	22.55	5.85	Returning
7	49.6	53.71	338.69	14.3445	76.5973	49.81	93.07	41.22	4.05	Returning
8	39.24	65.81	440.34	12.8604	77.6592	55.59	87.52	22.28	8.33	Searching
9	71.15	58.85	349.47	15.9993	76.1283	86.37	82.02	10.05	7	Idle
10	61.39	57.87	277.66	15.7115	79.5775	44.27	2.18	12.19	1.1	Searching

Module 2 - Data Preprocessing

Raw Data

Missing Values

Process Data

Before vs After

Variable	Missing_Values	Percentage
Drone_ID	0	0.00
Battery	0	0.00
Speed	0	0.00
Altitude	0	0.00
Latitude	0	0.00
Longitude	0	0.00
Signal	0	0.00
Target_Distance	0	0.00
Obstacle_Distance	0	0.00
Reward	0	0.00
Status	0	0.00

Module 2 - Data Preprocessing

Raw Data

Missing Values

Process Data

Before vs After

Select Preprocessing Technique:

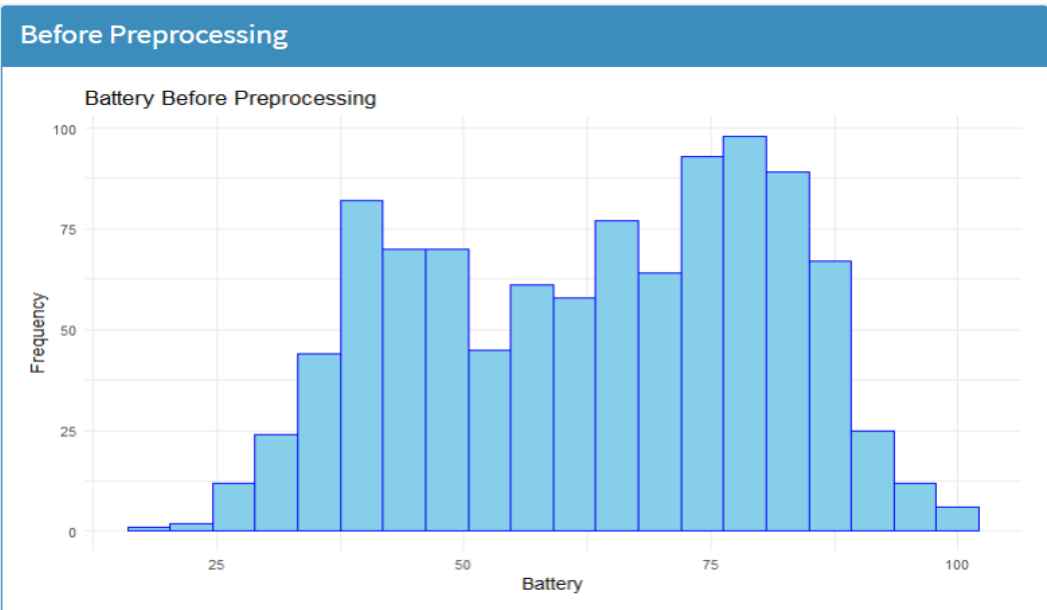
Min-Max Normalization

PROCESS DATA

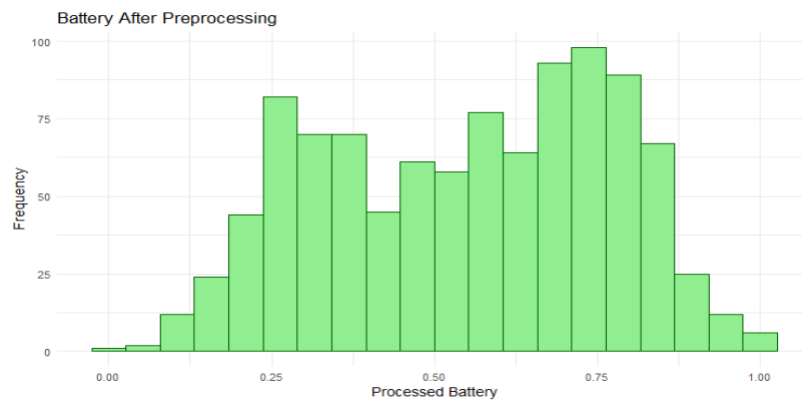
Show10entries

Search:

Drone_ID	Battery	Speed	Altitude	Latitude	Longitude	Signal	Target_Distance	Obstacle_D
All	All	All	All	All	All	All	All	All
0	0.6969474969474969	0.1213578088578089	0.2758572848983745	0.9320475967644894	0.5772629328344497	0.7980993664554852	0.3107701564380264	0.0869275
0.001001001001001001	0.3316239316239316	0.5718240093240093	0.7750132957199343	0.5918343472157228	0.4879983964454533	0.8792930976992331	0.324709185720016	0.412850
0.002002002002002002	0.4021978021978022	0.2738927738927739	0.4497421786482299	0.07913296343338451	0.741894532881221	0.2424141380460153	0.8720417168070598	0.09759893
0.003003003003003003	0.2606837606837607	0.6952214452214452	0.8002173561171873	0.8464970920516075	0.2241802662568698	0.5595198399466488	0.3289209787404733	0.9161849
0.004004004004004004	0.3428571428571428	0.7215909090909091	0.7328138367979282	0.02902934688147576	0.7486094175422193	0.9049683227742583	0.1253509827517048	0.0642507
0.005005005005005005	0.684004884004884	0.1841491841491842	0.2261428538395727	0.8160639080152416	0.8247615547797625	0.008002667555852017	0.3564981949458483	0.3897287
0.006006006006006006	0.3846153846153846	0.4663461538461539	0.5253312368487987	0.3909853599839559	0.09783353099370319	0.1633877959319774	0.9326113116726834	0.8048021
0.007007007007007007	0.2581196581196581	0.6426282051282051	0.7603764422965754	0.1429574169396349	0.2745419012143572	0.2597532510836947	0.8769554753309265	0.3837261



After Preprocessing



Module 3 - Multi-Agent Drone Swarm Simulation

Simulation Controls

Number of Drones:

Simulation Steps:

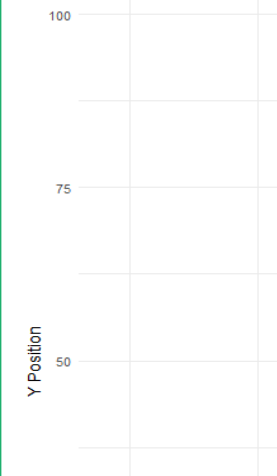
Target X Position:

Target Y Position:

START SIMULATION

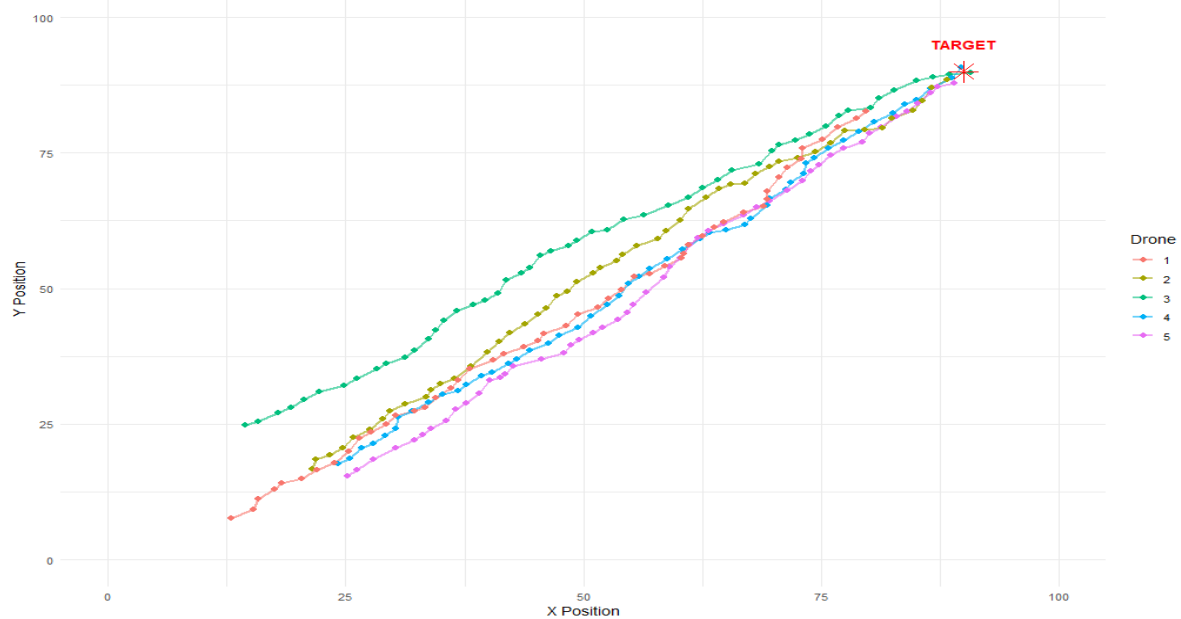
Drone Swarm Movement

Cooperative Drone Sw



Drone Swarm Movement

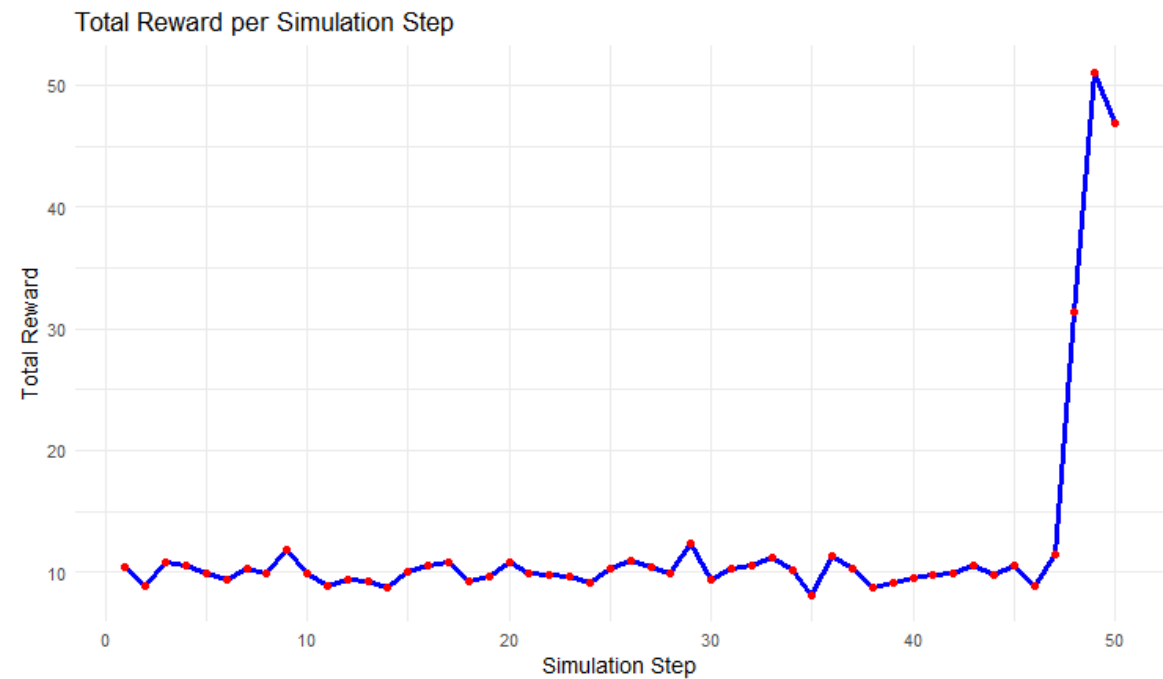
Cooperative Drone Swarm Movement



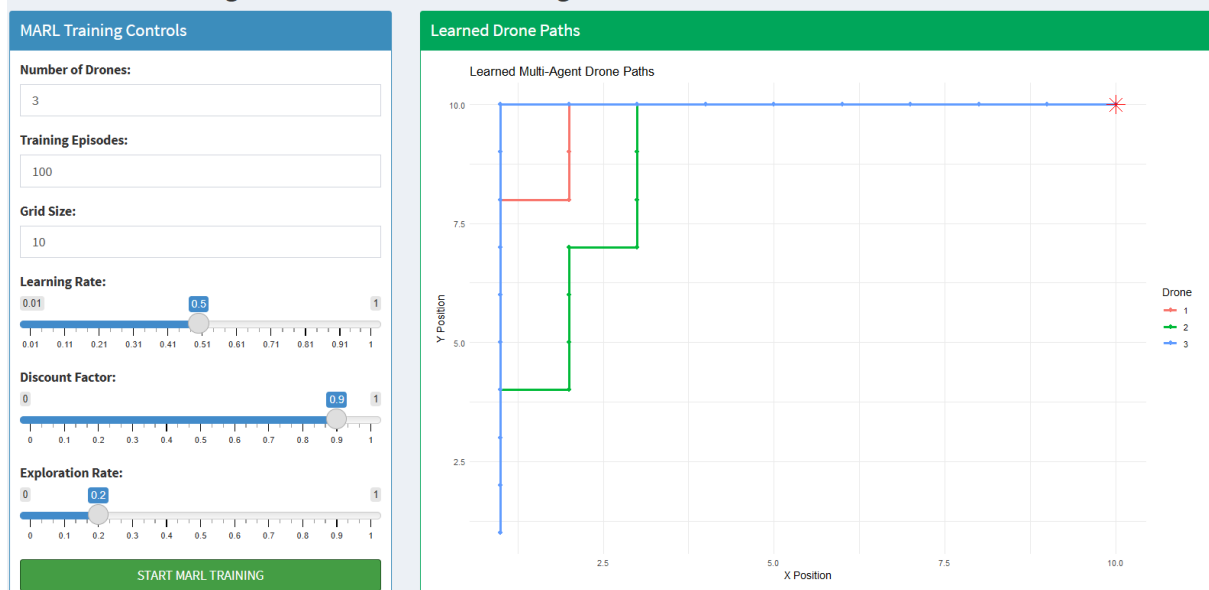
Simulation Results

Number_of_Drones	Simulation_Steps	Average_Final_Distance	Total_Reward
5	50	3.77	601.12

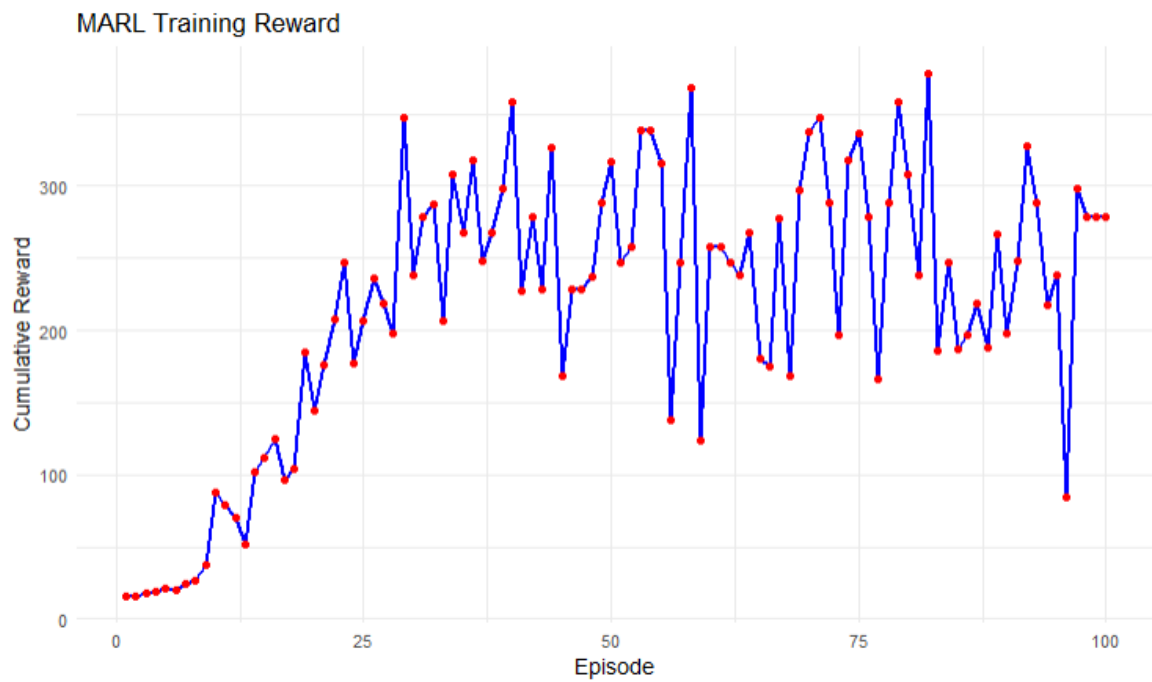
Reward Analysis



Module 4 - Multi-Agent Reinforcement Learning



Training Reward



MARL Results

Number_of_Drones	Training_Episodes	Average_Reward	Maximum_Reward	Final_Rev
3	100	217.86	378.18	27

Module 5 - Performance Evaluation

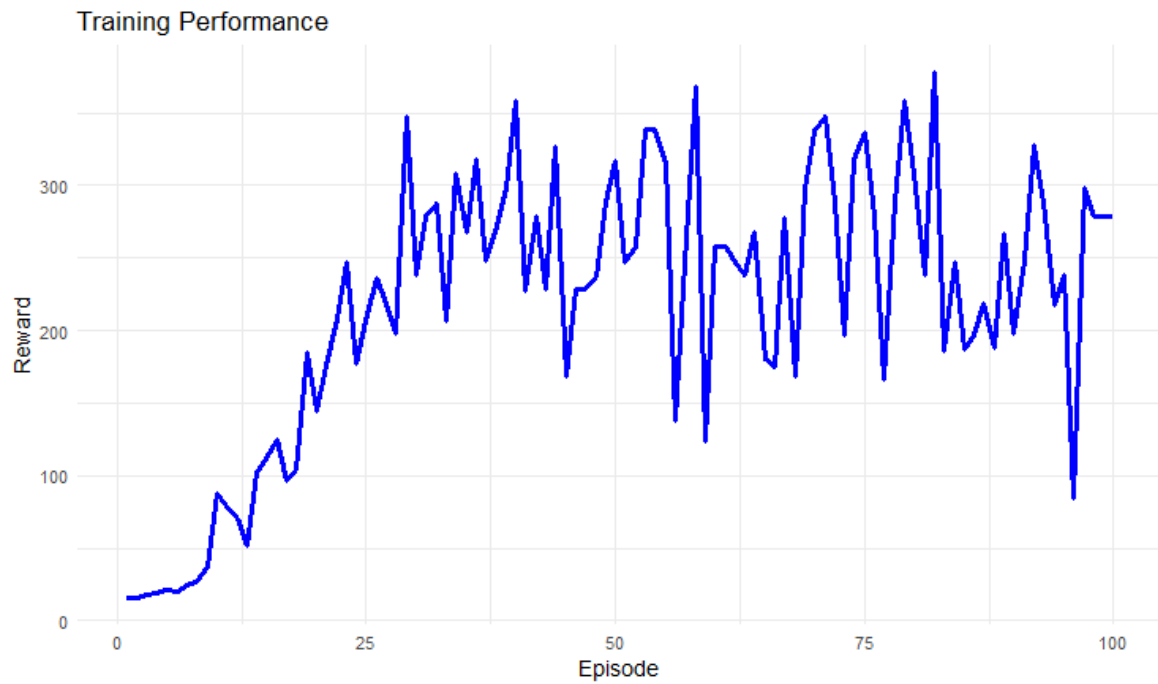
Evaluation

EVALUATE PERFORMANCE

Performance Metrics

Metric	Value
Average Reward	217.86
Maximum Reward	378.18
Average Steps	15.50
Final Distance	0.00
Success Rate (%)	100.00

Reward Performance



Performance Comparison

